



Multiple Choice:

1. Find the area bounded by the curve $r = a \cos 2\theta$.
 - A. $\frac{\pi}{2}a^2$
 - B. πa^2
 - C. $\frac{2}{3}\pi a^2$
 - D. $\frac{4}{5}\pi a^2$
2. What is moment with respect to the $x = 5$ axis of the area bounded by the curves $y = x^2$ from $x = 1$ to $x = 2$?
 - A. 9.72
 - B. 5.32
 - C. 7.92
 - D. 6.87
3. Find the area bounded by the parabola $x^2 = -y + 2$ and the line $x = y$.
 - A. 7.5
 - B. 2.3
 - C. 4.5
 - D. 6.5
4. Find surface area generated by revolving the curve $a^2y = x^3$ about the x -axis from $x = 0$ to $x = a$.
 - A. $4.45a^2$
 - B. $1.19a^2$
 - C. $7.89a^2$
 - D. $3.56a^2$
5. Evaluate $\int x \sin x dx$.
 - A. $-x \cos x + \sin x$
 - B. $-x \cos x + \cos x$
 - C. $-x \sin x + \sin x$
 - D. $-x \sin x + \cos x$
6. Integrate $\int \sin^3 x dx$.
 - A. $-\cos x + \frac{1}{3}\cos^3 x + C$
 - B. $-\cos x - \frac{1}{3}\cos^3 x + C$
 - C. $\cos x + \frac{1}{3}\cos^3 x + C$
 - D. $\cos x - \frac{1}{3}\cos^3 x + C$
7. If the region bounded by the lines $y = -\frac{4}{3}x + 4$, $x = 0$ and $y = 0$ is rotated about the y -axis, find the volume formed?
 - A. 18.8
 - B. 56.5
 - C. 37.7
 - D. 84.8
8. Evaluate $\int \frac{\sin \theta}{\cos^2 \theta} d\theta$.
 - A. $\sec \theta + C$
 - B. $\sin \theta + C$
 - C. $\csc \theta + C$
 - D. $\cot \theta + C$
9. Evaluate $\int_0^1 \frac{x dx}{(x+1)^8}$.
 - A. 0.011
 - B. 0.022
 - C. 0.033
 - D. 0.044
10. Find the moment of inertia, with respect to x -axis of the area bounded by the parabola $y^2 = 4x$ and the line $x = 1$.
 - A. 2.03
 - B. 2.13
 - C. 2.33
 - D. 2.53
11. Evaluate $\int_5^6 x(x-5)^{12} dx$.
 - A. 0.456
 - B. 0.556
 - C. 0.656
 - D. 0.756
12. Evaluate the integral of $dx / (x + 2)$ from -6 to -10 .
 - A. $2^{1/2}$
 - B. $1/2$
 - C. $\ln 3$
 - D. $\ln 2$
13. What is the integral of $\sin^5 x \cos^3 x dx$ if the lower limit is zero and the upper limit is $\pi/2$?
 - A. 0.0203
 - B. 0.0307
 - C. 0.0417
 - D. 0.0543
14. Evaluate the integral of $\ln x dx$, the limits are 1 and e .
 - A. 0
 - B. 1
 - C. 2
 - D. 3
15. Evaluate the double integral of $r \sin u dr du$, the limits of r is 0 and $\cos u$ and the limits of u are 0 and π .
 - A. 1
 - B. $1/2$
 - C. 0
 - D. $1/3$
16. Find the area in the first quadrant bounded by the parabola $y^2 = 4x$, $x = 1$ & $x = 3$
 - A. 9.555
 - B. 9.955
 - C. 5.955
 - D. 5.595
17. Locate the centroid of the plane area bounded by the equation $y^2 = 4x$, $x = 1$ and the x -axis on the first quadrant.
 - A. $(3/4, 3/5)$
 - B. $(3/5, 3/4)$
 - C. $(3/5, 3/5)$
 - D. $(3/5, 2/3)$



18. A body is moving horizontally under the action of a horizontal force $3x$ N, where x is the distance in meters of the body from the starting point, O. Find the work done in moving the body 8 m from O to A?

A. 24 J
B. 48 J
C. 96 J
D. 83 J

19. A force of 750 N compresses a spring 3 cm from its natural length of 15 cm. Find the work done in compressing the spring an additional 3 cm.

A. 3375 N-cm
B. 4500 N-cm
C. 1125 N-cm
D. 2250 N-cm

20. What is the surface area generated by revolving the parabola $y = x^2$ from $x = 0$ to $x = \sqrt{2}$ about the y-axis?

A. 12.74
B. 11.65
C. 14.98
D. 13.61

21. Evaluate $\int_0^2 x\sqrt{4x^2 + 9} dx$

A. 3.35
B. 4.57
C. 5.33
D. 8.17

22. A cable with mass $\frac{1}{2}$ kg/m is lifting a load of 150 kg that is initially at the bottom of a 50 m shaft. How much work is required to lift the load $\frac{1}{4}$ of the way to the shaft?

A. 20.804 kJ
B. 21.055 kJ
C. 22.810 kJ
D. 23.475 kJ

23. Evaluate

$$\iiint_B 8xyz \, dV, \quad B = [2, 3] \times [1, 2] \times [0, 1]$$

A. 12
B. 13
C. 14
D. 15

24. The volume of the solid obtained by revolving the region enclosed by the ellipse $x^2 + 9y^2 = 9$ about the x-axis is _____.

A. 2π
B. 4π
C. 6π
D. 9π

25. Find the average value of $1/x$ on the closed interval $[1, 3]$.

A. $1/2$
B. $2/3$
C. $\frac{1}{2} \ln 2$
D. $\frac{1}{2} \ln 3$

END